

Foundation (Jalapeno)

Q1. Solve $|x| = 5$

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve $|2x - 3| = 7$

- (A) $x = 2$ or $x = -5$
- (B) $x = 5$ or $x = -2$
- (C) $x = 1$ or $x = 4$
- (D) $x = 3$ or $x = -1$
- (E) $x = 0$ or $x = 7$

Q3. In a soccer game, the score can be represented by $|x - 2|$, where x is the number of goals scored by a team. If the score is 5, what are the possible values of x ?

- (A) $x = 7$ or $x = -3$
- (B) $x = 5$ or $x = -1$
- (C) $x = 3$ or $x = 1$
- (D) $x = 9$ or $x = -5$
- (E) $x = 0$ or $x = 10$

Q4. Solve $|3x + 2| = 11$

- (A) $x = 3$ or $x = -5$
- (B) $x = 2$ or $x = -4$
- (C) $x = 1$ or $x = -3$
- (D) $x = 4$ or $x = -2$
- (E) $x = 0$ or $x = 6$

Q5. A marathon runner's pace can be represented by $|x - 8|$,

where x is the runner's speed in miles per hour. If the pace is 3, what are the possible values of x ?

- (A) $x = 5$ or $x = 11$
- (B) $x = 4$ or $x = 12$
- (C) $x = 6$ or $x = 10$
- (D) $x = 7$ or $x = 9$
- (E) $x = 0$ or $x = 16$

Q6. Solve $|x + 4| = 9$

- (A) $x = 5$ or $x = -13$
- (B) $x = 3$ or $x = -11$
- (C) $x = 1$ or $x = -9$
- (D) $x = 4$ or $x = -8$
- (E) $x = 0$ or $x = 10$

Q7. Solve $|2x - 5| = 3$

- (A) $x = 1$ or $x = 4$
- (B) $x = 2$ or $x = 3$
- (C) $x = 3$ or $x = 1$
- (D) $x = 4$ or $x = 2$
- (E) $x = 0$ or $x = 5$

Q8. In a basketball game, a player's free throw percentage can be represented by $|x - 80|$, where x is the player's free throw percentage. If the free throw percentage is 10, what are the possible values of x ?

- (A) $x = 70$ or $x = 90$
- (B) $x = 60$ or $x = 100$
- (C) $x = 80$ or $x = 80$
- (D) $x = 85$ or $x = 75$
- (E) $x = 0$ or $x = 160$

Open Ended Questions (FRQ)

Q9. Solve $|x^2 - 4| = 5$ and check for extraneous solutions

Q10. A cyclist's distance from a starting point can be represented by $|x + 2|$, where x is the distance traveled by the cyclist. If the distance is 8, find the possible values of x and check for extraneous solutions

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to solve equations
- Consider multiple possible solutions for absolute value equations